

DUAL-MOTOR PMSM DYNAMOMETER PLATFORM

PLATFORM, ELECTRICAL & INTERFACES

TI LAUNCHXL-F28069M, dual BOOSTXL-DRV8305EVM stages, encoders, CAN, power, and physical verification

ENGINEERING REQUIREMENTS DOCUMENT | REVISION 2.4 | 5 AUGUST 2026

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Document scope	Physical platform, power distribution, motor phases, encoder routing, board stacking, CAN wiring, connectors, and release limitations
Evidence basis	Vendor schematics, connector maps, public BOM, board-interface figures, and setup verification records
Distribution	Customer-facing; selected source models and controlled engineering evidence included

DOCUMENT PURPOSE

Defines the controlled platform, electrical, interface, safety, verification, and acceptance requirements. Embedded algorithms, host software, CAN message handling, custom ADC code, and efficiency results are controlled by the companion Firmware & Host FRD.

Functional Requirements Document Framework

WHY THIS IS AN FRD

This document defines the physical and electrical behavior the platform shall provide, the interfaces and constraints under which it shall operate, and the objective evidence required for acceptance. Vendor diagrams and photographs support verification; they do not transfer ownership of third-party board designs.

Purpose, Scope, and System Boundary

This FRD governs the institution-independent public description of the dynamometer platform: DC input, board stacking, motor phases, encoder separation, CAN physical layer, deployment connection, mechanical restraint, and safe bench operation. Control algorithms and measured efficiency results are intentionally absent.

Boundary element	Included here	Excluded / referenced
Controller stack	LaunchPad and two inverter stages; connector compatibility	Vendor PCB manufacturing ownership
Machines	Two coupled encoder-equipped PMSMs and phase/feedback routing	Motor internal design and certified ratings
Bench interfaces	24 V input, USB/deployment, CANH/CANL/GND, termination	Host algorithms and message decoding

1. Platform Architecture

The platform uses one TI LAUNCHXL-F28069M controller board and two BOOSTXL-DRV8305EVM three-phase inverter stages. Two encoder-equipped PMSMs are mechanically coupled so one machine can drive while the second applies a controlled load. A current-limited 24 V bench supply powers the inverter stack; USB supports deployment and diagnostics; CAN supports supervisory communication.

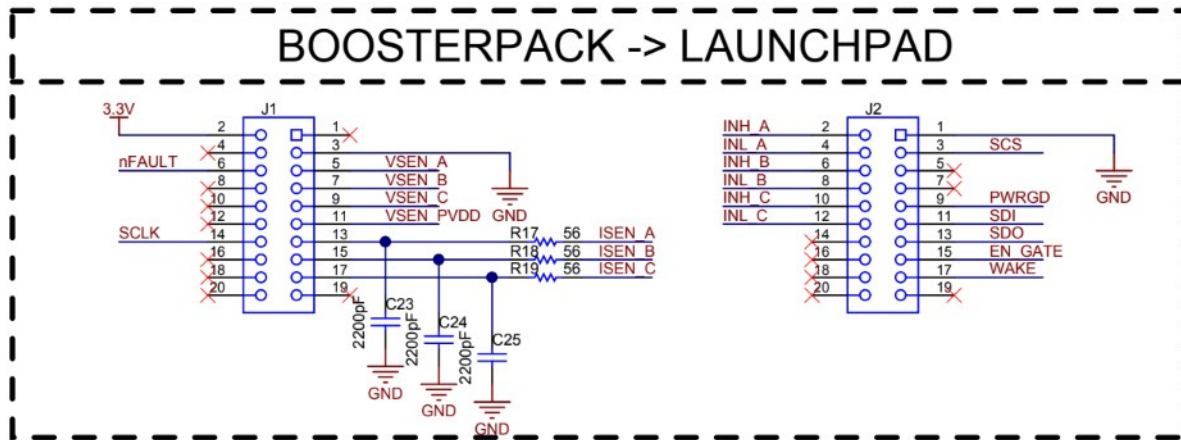


Figure 1. BoosterPack-to-LaunchPad board-interface arrangement.

1.1 Primary Elements

Element	Quantity	Public role
LAUNCHXL-F28069M	1	Embedded controller, encoder interfaces, PWM/ADC and CAN endpoint
BOOSTXL-DRV8305EVM	2	Three-phase inverter stages for drive and load machines
Encoder-equipped PMSM	2	Mechanically coupled drive and load machines
24 V DC supply	1	Current-limited bench power source
CAN interface	1	Supervisory physical-layer connection

2. Platform and Electrical Requirements

ID	The system shall...	Verification	Status
PLT-001	The platform shall provide two mechanically coupled encoder-equipped PMSMs.	Inspection	Demonstrated
PLT-002	The controller stack shall support two independent three-phase inverter stages.	Inspection	Demonstrated
PLT-003	The platform shall retain physical access for deployment, diagnostics, and emergency shutdown.	Inspection	Demonstrated
PLT-004	Rotating components shall be restrained and kept clear of hands during operation.	Procedure review	Required
PLT-005	The public release shall distinguish institution-owned or third-party hardware from authored models and source.	Document review	Verified

2.1 Power and Motor Interface Requirements

ID	The system shall...	Verification	Status
PWR-001	The inverter stack shall be supplied from 24 V DC with verified polarity and current limiting before operation.	DMM + startup test	Demonstrated
PWR-002	Each motor shall be connected consistently to its designated three-phase output.	Continuity inspection	Demonstrated
PWR-003	Initial power-up shall use conservative speed, current, and load limits.	Procedure review	Required
PWR-004	Software disable shall not be represented as an independent certified safety function.	Document review	Verified

3. Encoder, CAN, and Deployment Interfaces

ID	The system shall...	Verification	Status
IF-001	Motor 1 and Motor 2 encoder channels shall remain electrically and logically distinct.	Pin inspection + feedback test	Demonstrated
IF-002	CANH, CANL, and reference ground shall be connected consistently across the bus.	Inspection + bus test	Demonstrated
IF-003	The CAN bus shall provide two 120-ohm terminations at its physical ends.	Resistance inspection	Required
IF-004	The target shall retain the debug/deployment connection required for configuration and diagnostics.	Connection inspection	Demonstrated
IF-005	Board-revision pin assignments shall be checked against current vendor documentation before energizing hardware.	Document review	Required

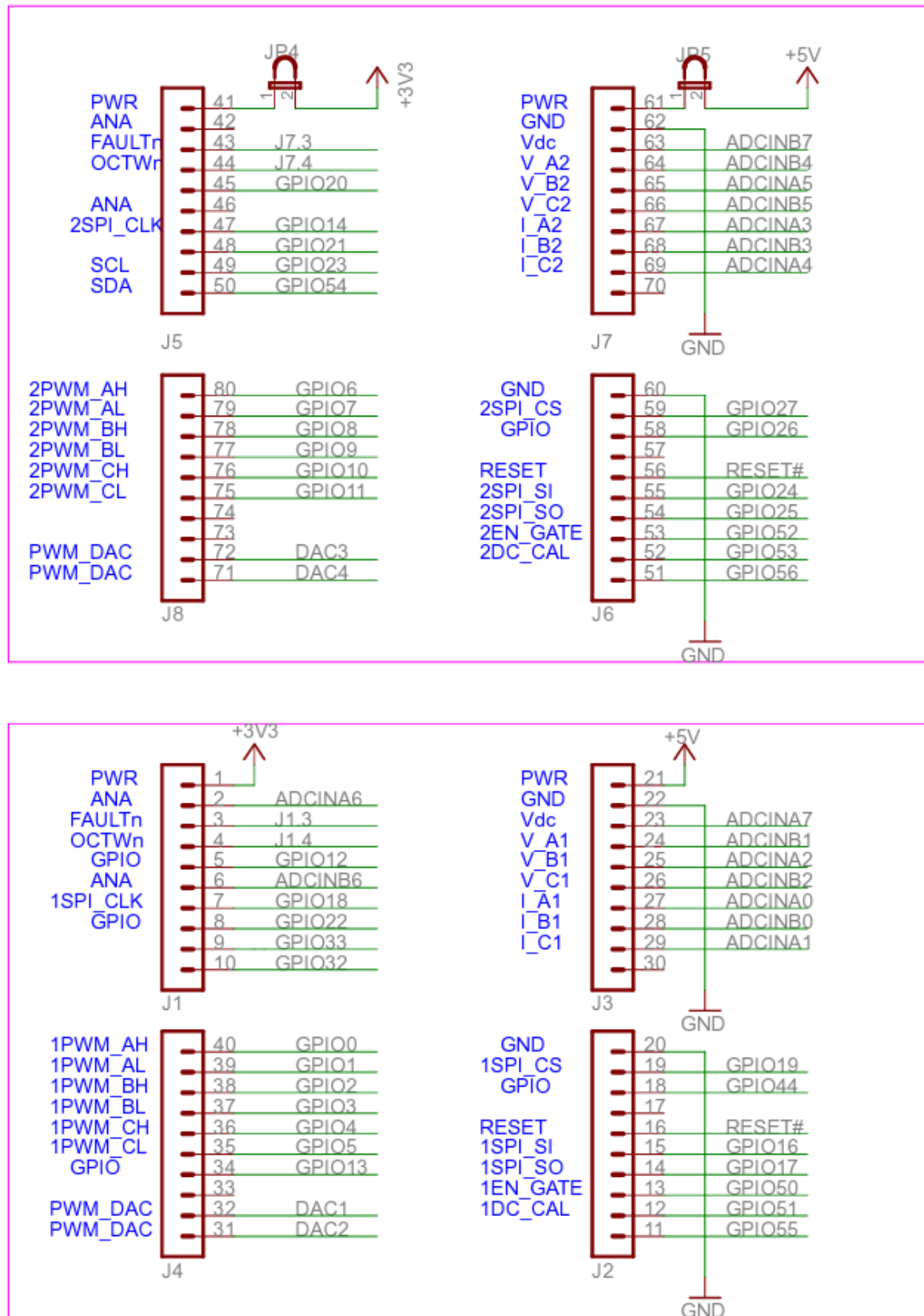


Figure 2. Public connector map used to verify board stacking and interface routing.

3.1 Interface Caution

BEFORE ENERGIZING

Verify supply polarity, board orientation, motor phase order, encoder routing, CAN termination, and mechanical restraint. Stop immediately if the shaft vibrates, loses tracking, overheats, or behaves unexpectedly.

4. Schematics, BOM, and Ownership Boundary

The public package includes vendor schematics and a high-level BOM for interface verification. These materials identify the equipment used; they do not claim authorship of TI board designs or transfer manufacturing rights.

Artifact	Public purpose	Boundary
LAUNCHXL schematic	Controller pin and peripheral reference	TI-owned reference material
BOOSTXL schematic	Gate-driver, phase-output and sensing reference	TI-owned reference material
Public BOM	Platform component identification	Not a manufacturing package
Connector maps	Wiring and board-stacking verification	Verify against current board revision

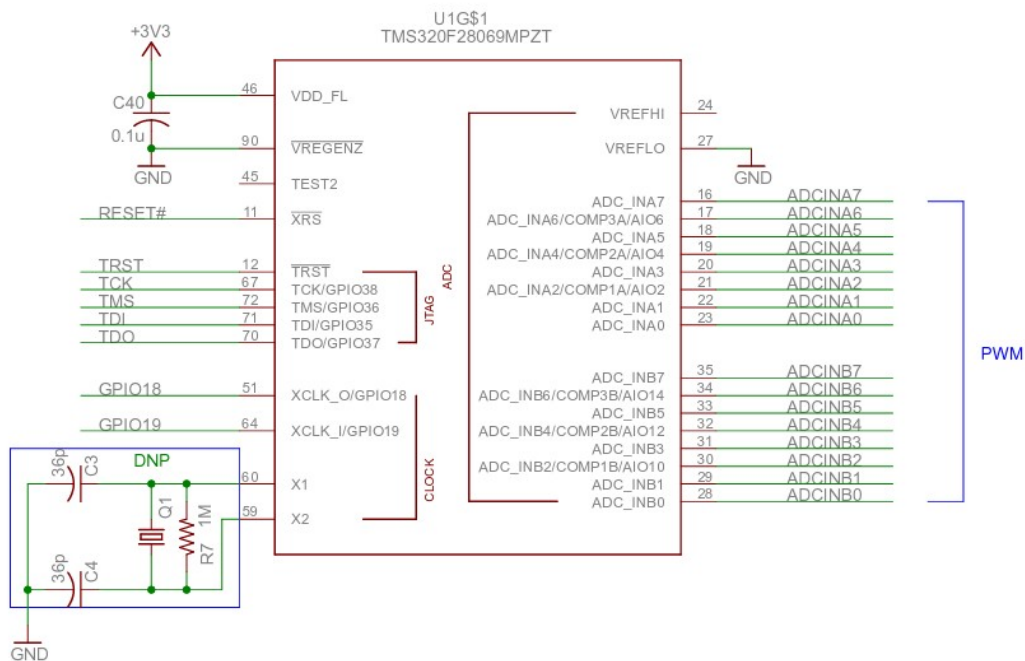


Figure 3. LaunchPad reference view retained for connector orientation.

5. Verification and Acceptance

Platform verification uses inspection, continuity checks, DMM polarity checks, current-limited startup, encoder feedback observation, and CAN communication tests. Passing document or CI checks does not replace physical verification.

Requirement group	Method	Evidence	Acceptance basis
PLT	Visual inspection	Board-interface figures and setup records	Required elements and physical access present
PWR	DMM + current-limited startup	Power-up procedure	Correct polarity and stable low-risk startup
IF	Continuity, pin inspection and bus test	Connector maps and communication evidence	Distinct encoder paths and working interfaces
Release	Archive inspection	NOTICE, BOM and schematics	Ownership and exclusions are explicit

5.1 Open Qualification Items

- Independent torque and electrical-power calibration are outside this release.
- Formal EMC, vibration, ingress, environmental, and long-duration thermal qualification are not included.
- A complete independent protection chain and certified stop architecture are not claimed.
- No vendor PCB manufacturing sources, Gerbers, pick-and-place files, or fabrication packages are redistributed.

5.2 Release Gate

ACCEPTANCE

Release is accepted when the interface requirements have defined verification methods, the schematics and connector maps correspond to the retained platform, the hardware ownership boundary is explicit, and the rendered FRD contains no firmware or efficiency-result duplication.